

SONIC EARTH ATLAS

“Hear Where Earth Is Changing.”

An Interactive Spatial Auditory Map Powered by Geographically Distributed NASA Earth Observations

1. EXECUTIVE SUMMARY & CORE INNOVATION

Sonic Earth Atlas is an interactive planetary exploration instrument designed for the 2026 NASA Space Apps Challenge (Earth Information Jukebox category). It is **not** a generic music visualizer and **not** an arbitrary "data to MIDI melody" converter.

Instead, the project establishes a **deterministic scientific sonification framework**: numerical NASA and NOAA satellite observations are processed through explicit normalization algorithms and synthesized into spatialized physical audio events. A user can intuitively explore Earth, perceive **WHERE** environmental events are happening via spatial audio positioning, identify **WHAT** is occurring through physically modeled sonic textures, and perceive **HOW IT CHANGES** over time.

1. WHERE (Spatial Audio)

Longitude maps directly to stereo panning: `pan = clamp(lon / 180, -1, 1)`. Latitude maps to acoustic elevation filters and HRTF 3D spatialization.

2. WHAT (Phenomenon Identity)

Physical synthesis signatures: Thermal combustion crackles for wildfires, downward droplet impacts for rainfall, and resonant pad drones for ocean anomalies.

3. HOW IT CHANGES (Temporal)

Timeline controls with a dedicated "Hear The Change" differential mode translating rates of change ($\Delta v = v_t - v_{t-1}$) into pitch and density flux.

2. SCIENTIFIC DATASETS & MATHEMATICAL NORMALIZATION PIPELINE

To preserve scientific integrity, raw machine-readable data is never mapped from visualization pixel colors. Instead, authoritative numerical values flow through validated mathematical transformations:

Phenomenon	Dataset & Source	Scientific Variable	Range & Unit	Normalization Equation	Synthesis Character
Active Wildfires	NASA FIRMS / VIIRS (375m)	Fire Radiative Power (FRP)	5 to 800 MW	<code>norm = clamp((log10(FRP) - log10(5)) / (log10(800) - log10(5)), 0, 1)</code>	Combustion crackle bursts; Bandpass 800–3000 Hz; Rate 4–24 Hz
Precipitation Rate	NASA GPM IMERG Final Run	Surface Rain Rate	0.1 to 45 mm/hr	<code>norm = clamp(sqrt(RainRate / 45), 0, 1)</code>	Resonant droplet pings; 340–700 Hz; Ping rate 2–18 Hz
Sea Surface Temperature	NOAA / NASA PO.DAAC / GHRSST	SST Thermal Anomaly	-3.0 to +3.0 °C	<code>norm = clamp((Anomaly - (-3.0)) / (3.0 - (-3.0)), 0, 1)</code>	Harmonic pad drone centered at 110 Hz (A2) ± 4 semitones; 0.18 Hz LFO

3. SYSTEM ARCHITECTURE & MODULE MAP

The codebase follows a strictly decoupled, production-grade modular architecture (React 19 + TypeScript + Vite 8 + Three.js + Tailwind CSS):

Module / File	Responsibility
<code>src/audio/sonificationEngine.ts</code>	Central deterministic audio engine managing voice budget (max 12 simultaneous voices), event prioritization, and layer routing.
<code>src/audio/fireSynthesizer.ts</code>	Deterministic pink/crackle buffer + bandpass filter + dynamic pulse interval rescheduling for active wildfires.
<code>src/audio/rainSynthesizer.ts</code>	Dual-oscillator droplet ping generator with exponential decay envelopes and dynamic pulse interval rescheduling.
<code>src/audio/oceanSynthesizer.ts</code>	Multi-oscillator pad with low-pass filter sweeps, glissando pitch modulation, and slow swell LFO.
<code>src/audio/spatialPanner.ts</code>	Web Audio HRTF <code>PannerNode</code> and <code>StereoPannerNode</code> chain with elevation filters.
<code>src/globe/GlobeCanvas.tsx</code>	3D WebGL Earth with procedural continental texture, atmospheric glow shader, interactive observation markers, and radiating acoustic sound rings.
<code>src/map/Accessible2DMap.tsx</code>	Equirectangular 2D accessibility-first map with full keyboard navigation (Arrow keys + Enter/Space) and audio crosshair.
<code>src/accessibility/AudioFirstMode.tsx</code>	Screen-reader accessible tabular console with live Web Speech API narration and high-contrast data display.
<code>src/datasets/adapters.ts</code>	Adapter pattern managing verified 6-timestep NASA/NOAA observational baseline with open meteorological reanalysis live proxy bridge.
<code>src/demo/GuidedDemoModal.tsx</code>	35-second choreographed planetary tour with synchronized captions and layer sequencing.

4. CORE INTERACTION & USER EXPERIENCE FEATURES

- **Opening Landing Screen:** Clean hero experience with primary CTA [LISTEN TO EARTH] (which initializes the Web Audio API strictly upon user gesture) and secondary CTA [EXPLORE THE GLOBE].
- **Auditory Legend:** Features dedicated [Hear Fire], [Hear Rain], and [Hear Ocean] audition buttons for first-time user familiarization.
- **Interactive Timeline & Temporal Flux:** Scrubber across 6 global timesteps with speed selectors (0.5x, 1x, 2x) and the [HEAR THE CHANGE] differential mode.
- **Location Inspector Modal:** Clicking any point displays exact scientific coordinates, measured value, normalized index, stereo pan position, and algorithmic confidence score.
- **Scientific Mapping & Provenance Panels:** Complete mathematical disclosures, NASA ESDIS / FIRMS / GPM attribution, and official repository links.
- **Real-Time System Diagnostics:** Real-time FPS counter, AudioContext state, sample rate, voice pool count (e.g. 8/12 voices), and visible observation count.

5. ACCESSIBILITY FIRST DESIGN

Accessibility is baked into the foundation rather than treated as an afterthought:

- **Full Keyboard Navigation:** Arrow keys move the audio focus crosshair across geographic coordinates; Enter/Space immediately triggers audio playback.
- **Audio-First Mode:** Provides a high-contrast tabular readout with automated voice narration (Web Speech API) for low-vision users.
- **Audio Safety & Protection:** Brick-wall dynamic compression limiter (-4 dB threshold, 16:1 ratio) prevents ear fatigue and digital clipping.
- **Stereo Fallback:** Seamless fallback if a user device does not support HRTF 3D audio.

6. AUTOMATED TESTING & VERIFICATION

An automated Vitest test suite (`src/__tests__/scientificSonification.test.ts`) guarantees algorithm correctness:

- ✓ Logarithmic decibel normalization for Fire FRP (5 to 800 MW → 0.0 to 1.0)
- ✓ Square-root scaling for GPM rain rate (0.1 to 45 mm/hr → 0.0 to 1.0)
- ✓ Bipolar zero-centered normalization for SST Anomaly (-3.0 to +3.0 °C → 0.0 to 1.0, 0 °C = 0.5)
- ✓ Longitude to stereo pan clamping (-180° to +180° → -1.0 to +1.0)
- ✓ Latitude acoustic elevation tilt factor (-90° to +90° → -1.0 to +1.0)
- ✓ Multi-timestep dataset integrity and delta (Δ) calculation across 6 timesteps
- ✓ Zero-crash DatasetAdapter fallback behavior

7. PRIOR ART & NASA ATTRIBUTION STATEMENT

Sonification of environmental parameters has an established history across scientific research. This project does not claim that data sonification or 3D globes are our exclusive invention. Rather, **Sonic Earth Atlas** focuses on an integrated, deterministic, spatial audio instrument that unifies geographic coordinates (WHERE), phenomenon synthesis physics (WHAT), and temporal variance (HOW IT CHANGES) into an accessible multi-sensory Earth map.

Attribution: Data and imagery provided by NASA and its Earth Science Data Systems (ESDIS, FIRMS, GPM, PO.DAAC) and NOAA. Built for the NASA Space Apps Challenge 2026.

